

Cerebellum

Anatomy, function and microstructure

Computational Neuroscience

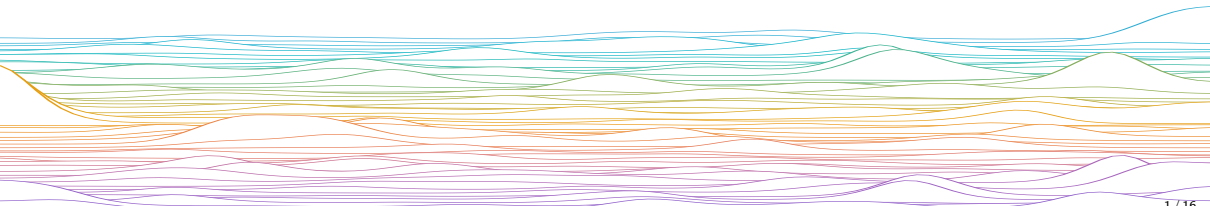
University of Bristol

Slides adapted from L Aitchison

M Rule

Learning outcomes:

- Cerebellar gross anatomy, microstructure, function, and the link to supervised learning



Cerebellum

“little brain”

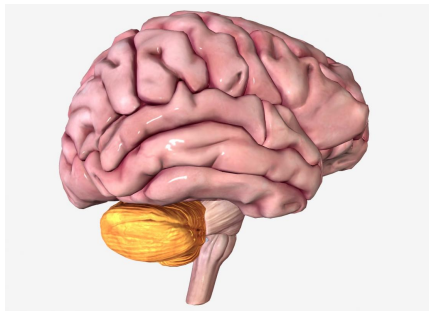
80% of the brain's neurons (Azevedo et al. 2009)

- ▶ of which most are cerebellar *granule cells*
- ▶ ~70 billion cerebellar granule cells (Lange 1975)

Damage → ataxia & cognitive deficits

Function:

- ▶ Forward modelling
 - Reshape motor commands from cortex to remove predictable errors
- ▶ Doya (1999,2000)
 - The Cerebellum is the brain's supervised learning module
- ▶ Cognitive roles
 - Ito (2008) *Control of mental activities by internal models in the cerebellum*



Cerebellar motor deficits

A standard test of cerebellar function is to reach with the tip of the finger for a target at arm's length: A healthy person will move the fingertip in a rapid straight trajectory, whereas a person with cerebellar damage will reach slowly and erratically, with many mid-course corrections.

Motor deficits with cerebellar damage:

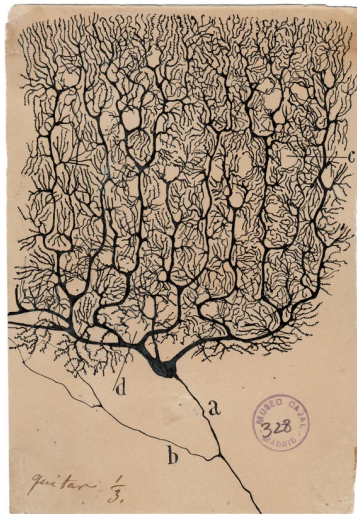
- ▶ speech deficits (slow speech, slurring)
- ▶ oculomotor deficits (e.g. instability of gaze)
- ▶ tremor
- ▶ difficulties in balance causing swaying + broad based stance



Cerebellar Ataxia, André Thomas (1012) Nervous and mental disease monograph series, issues 12, [via Wikimedia](#)

Cerebellar Purkinje cells (PCs)

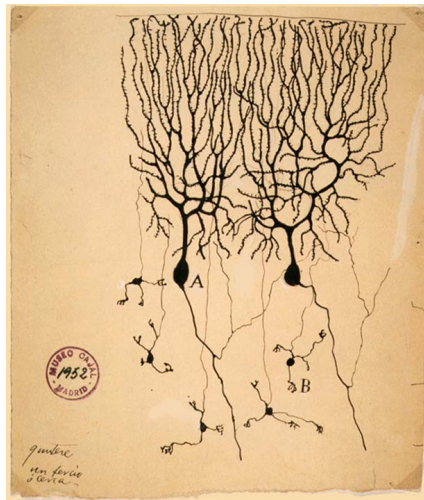
- ▶ Dendrites are planar!
- ▶ 100,000 input synapses (far more than most other cell types)
- ▶ Typical (simple-spike) firing rate = 50 Hz!



Cajal

Cerebellar Purkinje cells (PCs)

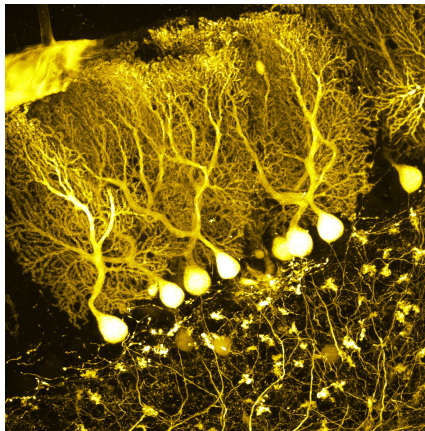
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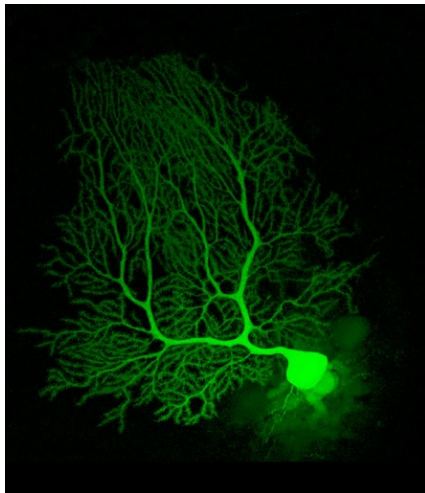
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[BrainsRusDC](#)

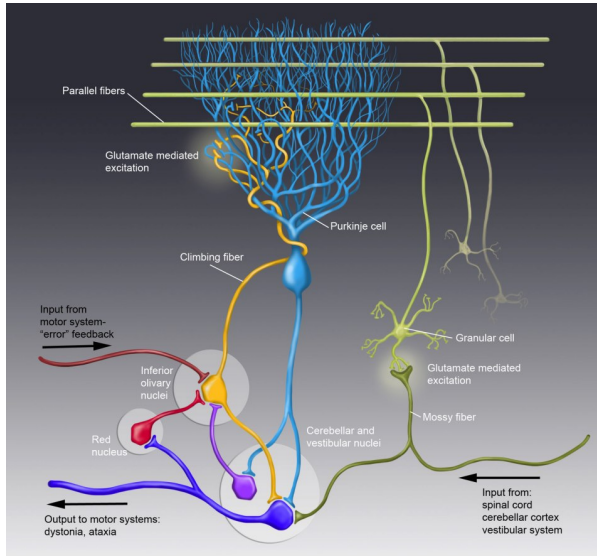
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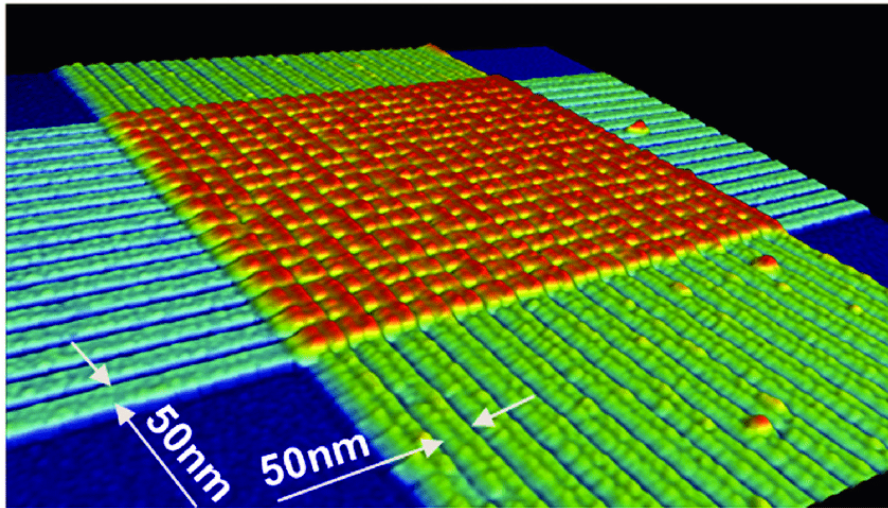


Maryann Martone, CCDB/NCMIR/UCSD

Cerebellar Microcircuit

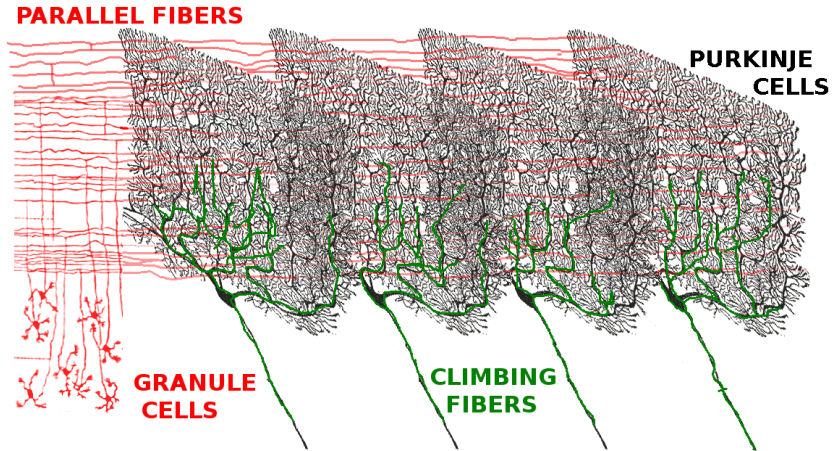


Cerebellar Microcircuit



memristor crossbar array

Cerebellar Microcircuit

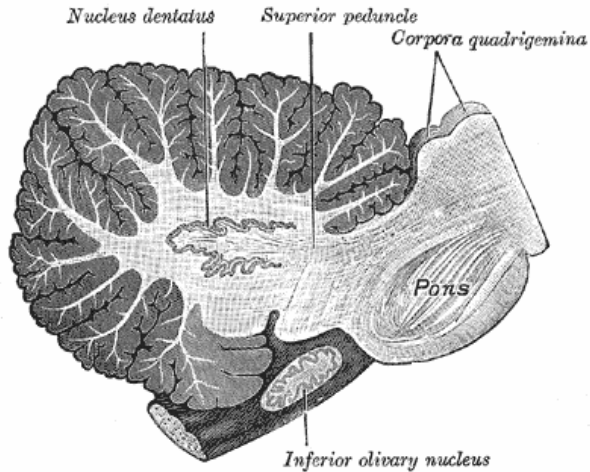


C Houghton

Inferior olive → climbing fibres

Pons (others) → mossy fibres

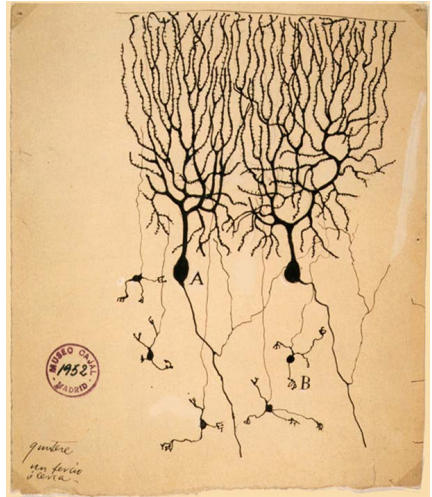
Purkinje cell output → dentate nucleus
(others)



Gray's anatomy

Granule cell inputs

- ▶ "B" in the image
- ▶ Most numerous cell type in the brain (70 billion cells)
- ▶ 1-7 synapses from mossy fibres → sparse, decorrelated representation
- ▶ Outputs → "parallel fibres"
- ▶ Granule cells themselves take input through the mossy fibres from all over the brain



Cajal

Climbing fibre error signals

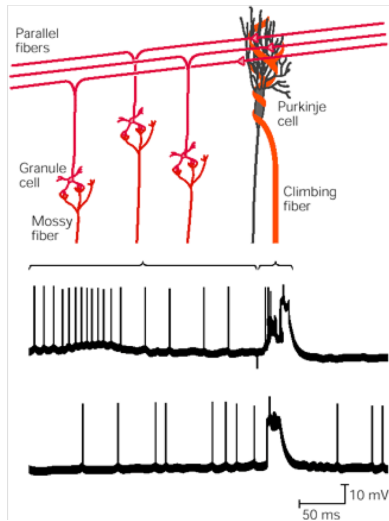
- ▶ There is one climbing fibre for each Purkinje cell.
- ▶ Climbing fibres wrap around the PC and synapse very, very strongly.
- ▶ Climbing fibres fire at a low rate ~ 1 Hz
- ▶ Climbing fibres originate in the inferior olive.



[Wilson et al\(2019\)](#)

Simple and complex spikes

- ▶ Standard PC inputs cause high-frequency (~50 Hz) simple-spikes.
- ▶ Climbing fibre inputs causes low-frequency complex spike (~1 Hz)
- ▶ Complex spikes signal errors, and cause LTD (i.e. indicate simple-spike output was too high)



Martinez et al. (1971)

Cerebellum is assumed to do supervised learning

Climbing fibre spikes

- ▶ Indicate a negative error
- ▶ Cause long-term depression of recently active granule→Purkinje synapses

$$\hat{y} = \sum_i w_i x_i$$

$\hat{y}$ Purkinje cell simple spikes

x_i Parallel fibre / granule-cell inputs

w_i granule-cell → Purkinje cell synapses

weight update:

$$\Delta w_i \propto \varepsilon \cdot x_i$$

$$\varepsilon = \underbrace{\min\{0, y^* - \hat{y}\}}$$

climbing fibre errors

If climbing fibres only cause LTD, what causes LTP?

- ▶ Calcium spikes in the dendrites (with sufficient input)
- ▶ Recall the *dendritic integration* slides, and view of neuron as $n \geq 2$ -layer net

Cerebellum self-test

The Cerebellum

- ▶ Contains ____ of the brain's neurons,
- ▶ Computes _____ models (motor and cognitive)
- ▶ Damage causes _____

The microcircuit

- ▶ _____ bring inputs from the brain to cerebellar _____
- ▶ _____ granule cells give rise to _____
- ▶ Purkinje cells have up to _____ parallel fibre inputs
- ▶ Each Purkinje cell received error feedback from a single _____ from the inferior olive

The cerebellum performs _____ learning

- ▶ Climbing fibre spikes signal a _____
- ▶ and cause long-term _____ of granule→Purkinje cell synapses

Cerebellum self-test

The Cerebellum

- ▶ Contains 80% of the brain's neurons,
- ▶ Computes forward models (motor and cognitive)
- ▶ Damage causes ataxia, various motor symptoms

The microcircuit

- ▶ Mossy fibres bring inputs from the brain to cerebellar granule cells
- ▶ ~70 billion granule cells give rise to parallel fibres
- ▶ Purkinje cells have up to 100,000 parallel fibre inputs
- ▶ Each Purkinje cell received error feedback from a single climbing fibre from the inferior olive

The cerebellum performs supervised learning

- ▶ Climbing fibre spikes signal a negative prediction error
- ▶ and cause long-term depression of granule→Purkinje cell synapses

end

